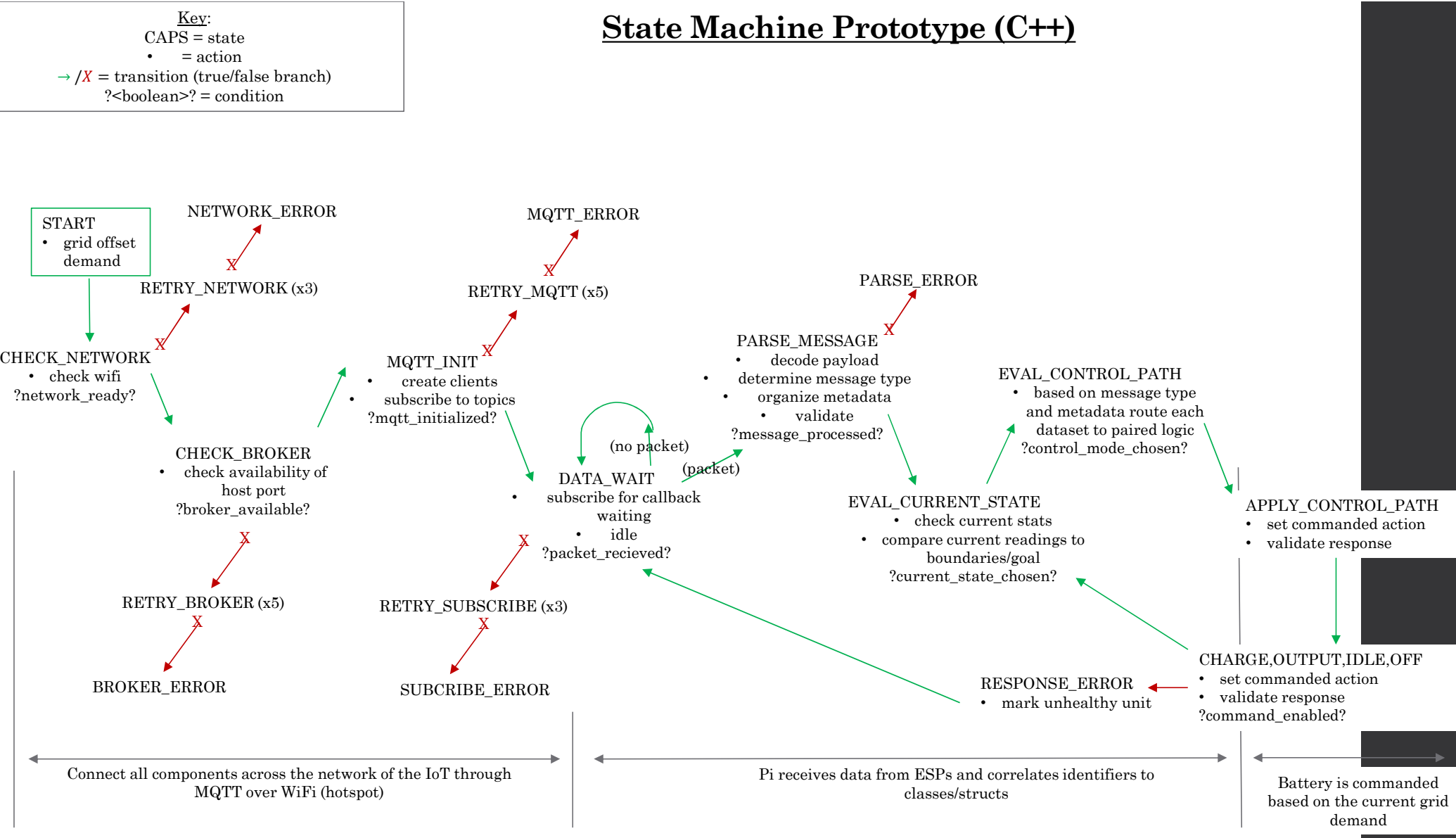


State Machine Prototype (C++)

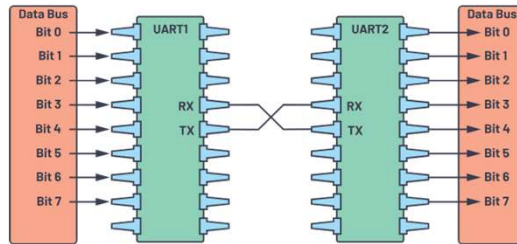


Class vs Struct Data

- AC/DC Output Commands
- Plug/Relay On/Off Commands
- Telemetry Reads
- Log Data
- Publish MQTT
- Connect
- Disconnect
- Poll Scheduling/Timing
- Checks
- Detect/Categorize Faults
- Retry Failed Connection
- User Human Readable Alerts
- Save/load Configuration

- Total Battery Percentage (SoC)
- AC/DC Output On
- AC/DC Input Power
- AC/DC Output Power
- Power Generation
- Plug Voltage (V)
- Plug Current (A)
- Plug Power (W)
- Plug Energy (kWh)
- Plug Frequency (Hz)
- MQTT Status
- Uptime
- Firmware Version
- Wifi-RSSI
- IP Address
- Serial
- Dsp Version
- Arm Version
- Error Flags
- Alarm Flags
- ESP32 Uptime
- ESP32 Temperature

Troubleshooting the ESP to Pi MQTT Bridge Using UART



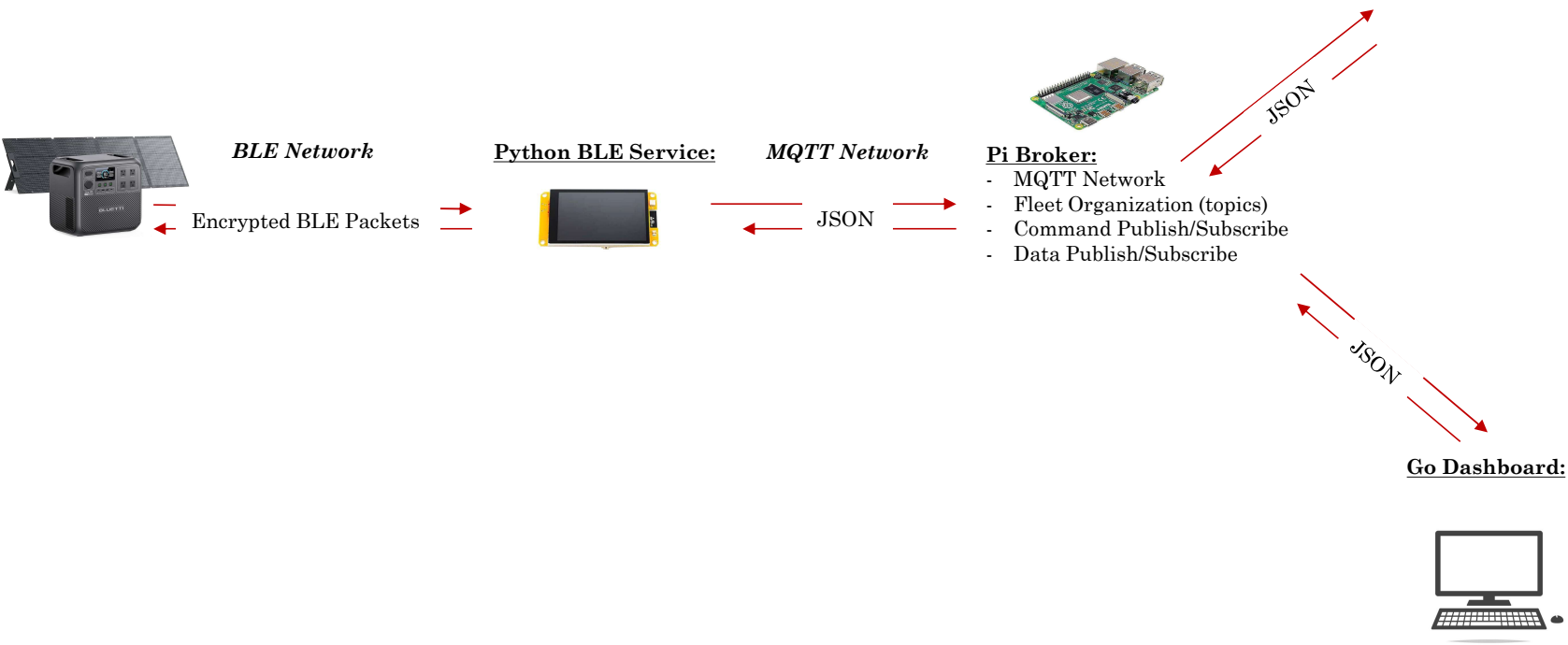
Command	Result
<pre>Serial.begin(<baud>); void loop(){ Serial.println("<string>");}</pre>	Establishes shared connection transmission speed and tells the UART peripheral to transmit data bytes

The screenshot shows a Windows terminal window with three panes: PROBLEMS, OUTPUT, and DEBUG CONSOLE. The OUTPUT pane displays the execution of a PlatformIO task: 'Executing task: C:\Users\HP\.platformio\penv\Scripts\platformio.exe device monitor'. The DEBUG CONSOLE pane shows the serial output of the device, including status messages like 'LVGL Setup done', 'deactivate relais contact', and 'Loading Values from EEPROM'. It also shows a large JSON object representing the device's configuration, including fields like 'plug', 'ft', 'tp', 'cmd', 'stat', 'tele', 'r1', 'swc', 'swn', 'btn', 'so', '14', 'lk', 'lt_st', and 'bat'.

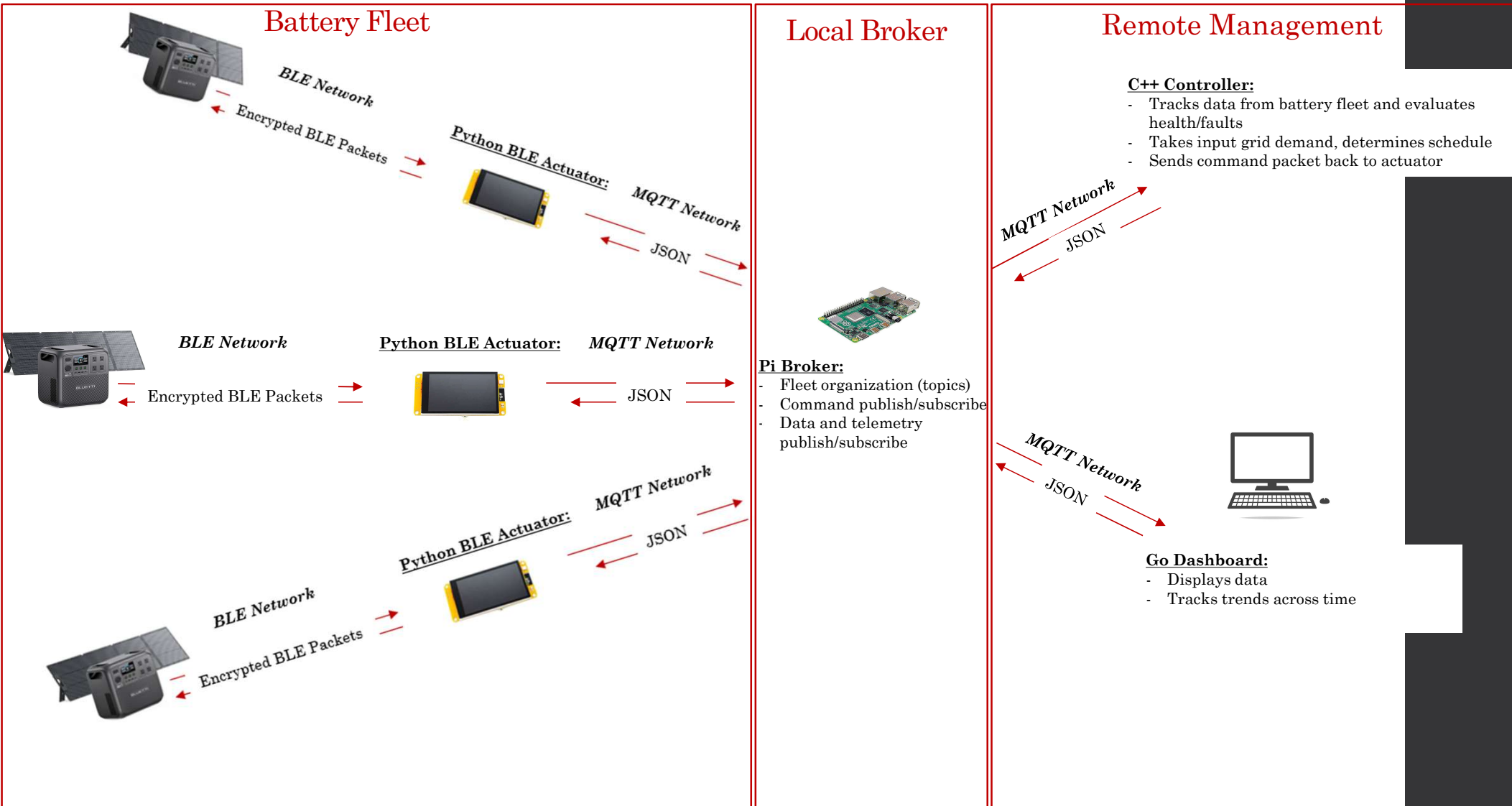
System Inter-Language Coms Prototype:

C++ Controller:

- config: Constants and parameterized inputs
- devices: Topic identification, variable initialization
- mqtt_client: Establish MQTT Client and functions
- controller: State machine, networking, battery health logic and faults
- debug: All prints for easy removal
- main: initialize the controller once at startup, then move into the main controller run loop



System Prototype:



JSON Parsing (Python, C++, and Go)

docs.python.org/3/library/json.html

3.14.6 Quick search Go

json — JSON encoder and decoder

Source code: [Lib/json/_init_.py](#)

github.com/nlohmann/json

nlohmann / json

Code Issues 29 Pull requests 58 Agents Discussions Actions Projects More

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JSON for Modern C++

MIT License

[json.nlohmann.me](#)

pkg.go.dev/encoding/json#example-Decoder

GO json package standard library

Version: go1.26.5 Latest Published: Jul 7, 2026

```
23 def json_publish_battery_state(state):
24     payload = {
25         "soc": state.soc,
26         "ac_output": state.ac_output,
27         "dc_output": state.dc_output,
28     }
29     mqtt_client.publish(
30         config.TOPIC_TELEMETRY,
31         json.dumps(payload)
32     )
33
34 def json_process_message(msg):
35     payload = json.loads(msg.payload)
36     return {
37         "soc": payload.get("soc"),
38         "ac_output": payload.get("ac_output"),
39         "charge_limit": payload.get("charge_limit"),
40     }
```

```
static DeviceInformation latestBattery; // temporary storage for the newest telemetry received
static void OnMessage(
    mosquitto*,
    void*,
    const mosquitto_message* msg){

    //display received topic for debugging
    std::cout<<"Topic: "<<msg->topic<<std::endl;

    //display received payload for debugging
    std::cout<<"Payload: "<<std::string(static_cast<char*>(msg->payload),msg->payloadlen)<<std::endl;
    //later:
    //assemble payload
    std::string payload(static_cast<char*>(msg->payload), msg->payloadlen);
    try{
        json data = json::parse(payload);
        std::cout << std::setw(4) << data << "\n\n";//serialize json

        //update storage container variables with new data
        latestBattery.SOC = data["soc"];
        latestBattery.batteryCurrent = data["current"];
    }

    catch(const json::exception& e){
        std::cerr << e.what() << std::endl;
    }
}
```

Types

type Decoder

```
type Decoder struct {
    // contains filtered or unexported fields
}
```

A Decoder reads and decodes JSON values from an input stream.

Successful Automated Package Delivery From Python to C++ Through JSON!

```
[READ] DC Input Switch: [0]
[Read AC Input Switch] 00 08 b9 e8 d3 0c e9 00 aa 3
3 2e 6f 9b a5 c7 b0 5b 30 fc 3b b3 83
Finished storing values inside variables
[READ] AC Input Switch: [0]
[Read Grid Charging Permission] 00 08 cc 05 ed bf b
9 47 c4 7d d4 10 16 0b c3 3b 19 d1 6e ch 25 c9
Finished storing values inside variables
[READ] Grid Charging Permission: [1]
[Read AC Input Setting] 00 08 42 50 a1 ad 0a 7c d4
1b 30 fc d5 a4 31 d9 74 98 9e 0c fc eb
Finished storing values inside variables
[READ] AC Input Setting: [0]
[Read Home Telemetry Block] 00 08 96 78 50 fe 68 76
dd c9 26 09 bb 25 0a 0a 0d fs 69 3d 27 f6
Finished storing values inside variables
[READ] Home Telemetry Block: [3986, 28, 97, 2, 1986
, 1986, 0, 1, 1, 0, 19525, 12340, 48, 0, 0, 0, 3470
0, 22612, 606, 0, 1, 0, 1, 648, 2578, 0, 0, 0, 0, 0
, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 84, 0, 0]
[Read AC/DC Load Info Block] 00 08 21 4a 94 ed 07 8
b ef 86 a7 e8 44 54 8b 4f 1e 1a 01 a9 f7 10
Finished storing values inside variables
[READ] AC/DC Load Info Block: [0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 84, 0, 0, 0
, 0, 0, 0, 0, 0, 1, 83, 1196, 9, 0, 0]
[Read PV Input Info Block] 00 08 4b 6c 6b 9b f8 2d
6e 84 70 be bf d1 a1 a0 12 51 d7 6d 90 bb
Finished storing values inside variables
[READ] PV Input Info Block: [0, 0, 95, 0, 0, 0, 0,
0, 0, 1]
[]

System has a fault
System has a fault
PS C:\ecolibrum\smart_plug_PI> .\BatteryController.exe
STARTING CONTROLLER
Connected to MQTT broker.
Subscribed to tele/battery/STATE
System is idle
System has a fault
System has a fault
System has a fault
System has a fault
Topic: tele/battery/STATE
Payload: {"device": "bluetti_ac200", "timestamp": 178596284
1.0800385, "action": {"request": "read", "target": null, "v
alue": null, "status": null, "state": "idle"}, "telemetry":
{"home_telemetry": {"soc": 97, "voltage": 399.2, "current"
: 0.1, "pv_watts": 0, "grid_watts": 0, "ac_watts": 0, "dc_w
atts": 0}, "dc_output": {"value": 0}, "ac_output": {"value"
: 1}, "dc_input": {"value": 0}, "ac_input": {"value": 0}, "
grid_permission": {"value": 1}, "ac_input_setting": {"value
": 0}}}
timestamp
JSON parse failed: [json.exception.type_error.302] type mus
t be string, but is null
Failed to parse battery telemetry.
System has a fault
System has a fault
System has a fault
System has a fault
System has a fault
System has a fault
PS C:\ecolibrum\smart_plug_PI>

{"device": "bluetti_ac200", "timestamp": 1785962897.434542, "action": {"r
equest": "read", "target": null, "value": null, "s
tatus": null, "state": "idle"}, "telemetry": {"hom
e_telemetry": {"soc": 97, "voltage": 399.2, "curren
t": 0.1, "pv_watts": 0, "grid_watts": 0, "ac_watts"
: 0, "dc_watts": 0}, "dc_output": {"value": 0},
"ac_output": {"value": 1}, "dc_input": {"value": 0
}, "ac_input": {"value": 0}, "grid_permission": {"
value": 1}, "ac_input_setting": {"value": 0}}}
tele/battery/STATE {"device": "bluetti_ac200", "ti
mestamp": 1785962111.246163, "action": {"request":
"read", "target": null, "value": null, "status":
null, "state": "idle"}, "telemetry": {"home telem
etry": {"soc": 97, "voltage": 399.2, "current": 0.0
, "pv_watts": 0, "grid_watts": 0, "ac_watts": 0, "
dc_watts": 0}, "dc_output": {"value": 0}, "ac outp
ut": {"value": 1}, "dc_input": {"value": 0}, "ac_i
nput": {"value": 0}, "grid_permission": {"value":
1}, "ac_input_setting": {"value": 0}}}
tele/battery/STATE {"device": "bluetti_ac200", "ti
mestamp": 1785962125.1193569, "action": {"request"
: "read", "target": null, "value": null, "status":
null, "state": "idle"}, "telemetry": {"home telem
etry": {"soc": 97, "voltage": 399.2, "current": 0.
1, "pv_watts": 0, "grid_watts": 0, "ac_watts": 0,
"dc_watts": 0}, "dc_output": {"value": 0}, "ac out
put": {"value": 1}, "dc_input": {"value": 0}, "ac_
input": {"value": 0}, "grid_permission": {"value":
1}, "ac_input_setting": {"value": 0}}}
tele/battery/STATE {"device": "bluetti_ac200", "ti
mestamp": 1785962139.0119865, "action": {"request"}}
```